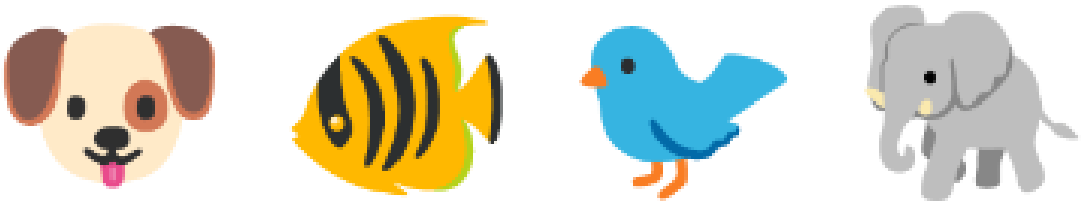
🐾 What Do ALL Animals Need? 🐾

Life Science — Animal Needs | Unit 3, Lesson 1 of 6

📚 **Grade:** Kindergarten ⏱️ **Duration:** 30–45 minutes 🔬 **Subject:** Life Science — Animal Needs
🌿 **Unit:** Unit 3, Lesson 1 of 6



From tiny fish to giant elephants — all animals share four basic needs to survive!

📋 **NGSS Standard:**
K-LS1-1 — Use observations to describe patterns of what plants and animals (including humans) need to survive.

🎯 Learning Objective

Students will **NAME and DESCRIBE** the four basic needs (food, water, air, and shelter) that all animals share, and give one example of each.

💡 The Big Idea

Every animal on Earth — from the tiniest ant to the biggest blue whale — needs four things to **survive**. These needs are the same no matter where an animal lives or what it looks like:

🍎 **Food**

💧 **Water**

🌬️ **Air**

🏠 **Shelter**

💡 **Key vocabulary:** survive, needs, food, water, air, shelter

📦 Materials

🏠 Gather These Household Items

- Bowl of water
- Small snack (crackers or raisins)
- A hand fan or piece of cardboard (to fan yourself and feel air — air is all around us, not stored in a container!)
- Small box or container (represents shelter)
- Stuffed animals or family pet
- Paper and crayons

🎲 Distractor Items for the Sort

- Toy car, crayon, coin, book, or any other household objects that are *not* survival needs

🔬 Let's Get Started

Step 1 — Introduce the Big Question (5 min)
Hold up a stuffed animal (or point to the family pet). Ask: *"What does this animal need to stay alive? What would happen if it didn't have food for a week? What about water?"*

Let kids brainstorm freely. Write or draw their ideas on paper. Then reveal: *"Scientists have figured out that ALL animals need exactly four things to survive. Let's find out what they are!"*

Step 2 — Meet the Four Needs (5–8 min)
Gather your four items and introduce them one at a time:

- 🍎 **Food** — Hold up the snack. *"Animals need food for energy. Without energy, they can't move, grow, or do anything!"*
- 💧 **Water** — Hold up the bowl. *"Every animal needs water. Even animals that live in deserts find ways to get water!"*
- 🌬️ **Air** — Fan yourself or breathe gently on your hand. Ask: *"Can you feel that? What is it?"* → Air! *"Air is all around us — you can't hold it in a box, but you can feel it. Animals need air — specifically a gas called oxygen — to breathe."*
- 🏠 **Shelter** — Hold up the small box. *"Shelter is a safe place to rest, stay warm, and hide from danger. It could be a den, a nest, a pond, or even a dog bed!"*

🌬️ **Air is everywhere!** Fan yourself with a piece of paper or a hand fan and ask "What do you feel?" (air / wind). Breathe gently on your child's hand — *"That's air! It's all around us — animals breathe it wherever they are, not from a container."* Great sensory hook for little learners.

Step 3 — Survival Sorting! (10–12 min)
Spread out all your items on the floor: the four survival items (snack, water bowl, hand fan/cardboard, box) AND the distractor items (toy car, crayon, coin, etc.).

Call out items one at a time and ask: *"Does a dog need this to survive?"*

- Bowl of water → ✅ YES — survival need (Water)
- Crackers → ✅ YES — survival need (Food)
- Fan/cardboard (wave it to feel air) → ✅ YES — survival need (Air — air is everywhere, we just used the fan to FEEL it!)
- Box → ✅ YES — survival need (Shelter)
- Toy car → ❌ Nope — fun, but not needed to survive!
- Crayon → ❌ Nope — same!
- Coin → ❌ Nope — animals don't use money 😄

Make two piles: **NEEDS** and **NOT NEEDS**.

Step 4 — Needs Hunt! (8–10 min)
Go on a "Needs Hunt" around the house! Look for evidence of where your pet (or stuffed animal) gets each need met:

- 🍎 **Food** — Where is the pet food bowl or food kept?
- 💧 **Water** — Where is the water bowl?
- 🌬️ **Air** — Open a window! Air is everywhere — animals breathe it wherever they are.
- 🏠 **Shelter** — Where does the pet sleep? Their bed, a tank, a cage?

If no pet: use a stuffed animal! *"Where would we put this animal's food bowl? Where would it sleep?"* Set it all up in a little pretend habitat corner.

Step 5 — Wrap-Up (3–5 min)
Point to each item on the floor and have kids shout out the four needs in order: *"FOOD! WATER! AIR! SHELTER!"*
Ask: *"Can you remember all four? Let's see if you can say them without looking!"*

💬 Discussion Question

🐠 vs 🐕 — Think about this together:

"Do you think a fish needs the same things as a dog?"

Give kids a moment to think and share their ideas before offering the answer below.

💡 **Suggested Answer:** Yes! Both a fish and a dog need food, water, air, and shelter — but they get those things in *different ways*.

A fish uses **gills** to pull the oxygen out of water — the same oxygen we breathe from air, just dissolved in water instead. A dog breathes air directly with **lungs** and might sleep in a cozy dog bed. Same four needs — very different methods!

Follow-up: "Can you think of another animal? What does IT need?"

🚀 Extensions

- 🎨 **Habitat Diorama:** Build a mini habitat for a stuffed animal in a shoebox. Include all four needs: a tiny food bowl, a water cup, a toy "shelter," and air (it's already there — just name it!).
- 📚 **Read Aloud:** *"What Do Animals Need?"* by Lisa J. Amstutz or *"Living Things and Their Needs"* by Daniel Nunn.
- 🌿 **Nature Walk:** Go outside and try to spot evidence of animal needs in the wild — a bird's nest (shelter!), a puddle (water!), berries on a bush (food!)
- 🔢 **Animal Matching Game:** Name an animal, name one of its four needs, and describe how it meets that specific need. Take turns!
- 📖 **Unit Preview:** *"Next time, we're going to look at just ONE need up close — stay tuned!"* (Builds anticipation for Lesson 2.)

👩‍🏫 Parent/Teacher Notes

- **No materials to buy** — everything needed is already in your home!
- **Introducing "air":** Fan your child's hand or breathe gently on it — ask "what do you feel?" Air is everywhere, not stored in a container. The key concept is that air is all around us; we don't need to capture it in something to show it exists.
- **No pet?** Use a stuffed animal as a stand-in. Set up a little "habitat corner" for it with a pretend food bowl, water cup, and bed.
- **Introduce vocab:** survive, needs, food, water, air, shelter. Keep coming back to these words throughout the lesson.
- **This is the foundation lesson** — all five other lessons in this unit build on these four needs. Getting kids fluent in "food, water, air, shelter" sets them up for everything that follows.
- **Don't worry about perfect sorting** — the discussion matters more than accuracy. If a child says "a toy car" is a need, ask them why! It's a window into their thinking.

🐾 My Animal Needs Worksheet 🐾

Life Science — Animal Needs | Unit 3, Lesson 1 of 6

Name: _____ Date: _____

In each box, draw one way your pet (or favorite animal) gets that need met!

🍎

Food

Draw how your animal gets its food!

💧

Water

Draw how your animal gets its water!

🌬️

Air

Draw how your animal breathes!

🏠

Shelter

Draw where your animal sleeps or hides!

🌍 **Bonus:** Can you name ALL FOUR needs without looking at the boxes above?

_____ _____ _____ _____

✂️ Survival Sorting Mat ✂️

Name: _____ Date: _____

Cut out the item cards below. Sort each one: does an animal NEED it to survive, or is it just extra?

✅ SURVIVAL NEED Animals need this to live!	❌ NOT A NEED Nice to have, but not required
Glue or place your NEED cards here	Glue or place your NOT A NEED cards here

✂️ Item Cards — Cut These Out!

Cut along the dashed lines and sort onto the mat above.

🍎 **Apples (food)**

💧 **Bowl of water**

🌬️ **Air (feel it — wave your hand!)**

🏠 **A cozy den (shelter)**

🚗 **Toy car**

🖍️ **Crayon**

🪙 **Coin**

🌿 **Grass (food for some!)**

🪺 **Bird's nest (shelter)**

📱 **Phone**

🐟 **Fish food (flakes)**

🎁 **Gift box**

📄 View Reference Card (Full Poster)

✅ PARENT ANSWER KEY — Detach before giving worksheet to students

✅ SURVIVAL NEEDS

🍎 Apples — *Food*

💧 Bowl of water — *Water*

🌬️ Air — *Air (oxygen) — it's all around us, we can feel it!*

🏠 A cozy den — *Shelter*

🌿 Grass — *Food (for herbivores like rabbits, horses, cows)*

🪺 Bird's nest — *Shelter*

🐟 Fish food (flakes) — *Food (what you feed a pet fish — a clear survival need!)*

❌ NOT SURVIVAL NEEDS

🚗 Toy car — fun for us, not for animals!

🖍️ Crayon — creative, but not survival

🪙 Coin — animals don't use money 😄

📱 Phone — no service in the wild!

🎁 Gift box — nice surprise, but not needed

💡 Discussion Hint Notes

- 🌿 **Grass card:** Kids might say grass isn't food — great teaching moment! It IS food for many animals (herbivores). Encourage kids to think about who eats what.
- 🪺 **Bird's nest:** Reinforce "shelter" — shelter doesn't have to be a house. It's any safe place to rest and stay protected. Nests, burrows, caves, and dens all count!
- 🌬️ **Air card:** Air is everywhere — reinforce that animals don't get air "from a container." Fan your hand or breathe on it: *"That's air! It's all around us — animals breathe it wherever they are."*
- **Fish and water:** Fish also need oxygen to survive — but instead of breathing air like we do, their gills extract dissolved oxygen directly from water. This connects to the Discussion Question about fish vs. dogs.

✅ What Success Looks Like

- Child can name all four needs: food, water, air, shelter (with prompting is fine at this stage!)
- Child can correctly sort at least 3 of the 4 survival need cards into the NEEDS column.
- Child can say one thing about how their pet (or favorite animal) gets at least ONE need met.
- Child can answer the discussion question: *"Yes, a fish and a dog need the same things — just differently!"*

📚 Unit Preview (For Your Planning)

This is **Lesson 1 of 6** in Unit 3: Animal Needs. Every lesson builds on these four needs:

- Lesson 2 — Food: What do different animals eat?
- Lesson 3 — Breathing: How do different animals breathe?
- Lesson 4 — How Animals Drink: How do animals find and use water?
- Lesson 5 — Building a Shelter: What makes a good home?

- Lesson 6 — When Needs Aren't Met: Putting it all together

🧠✅

Ready to test what you learned?

See how much you know about Animal Needs with the unit quiz!

→ Take the Unit Quiz

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🍽️ What's for Dinner? How Animals Eat 🍽️

Life Science — Animal Needs | Unit 3, Lesson 2 of 6

📚 **Grade:** Kindergarten ⏱️ **Duration:** 30–45 minutes 🌿 **Subject:** Life Science — Animal Needs 🔬 **NGSS:** K-LS1-1

Some animals eat plants. Some eat meat. Some eat both — just like us!

📋 **NGSS Standard:**
K-LS1-1 — Use observations to describe patterns of what plants and animals (including humans) need to survive.

🎯 Learning Objective

Students will **SORT at least 6 animals** into herbivore, carnivore, or omnivore categories and **EXPLAIN their reasoning** for at least one choice.

💡 Key Concept

Animals eat different things to survive! Some animals eat only plants, some eat only other animals, and some eat both. Scientists have special names for each group. Knowing what an animal eats helps us understand how it lives — and how all animals in nature depend on each other.

🐰

Herbivore Eats **plants only**
Leaves, grass, fruit, seeds

🦁

Carnivore Eats **meat only**
Other animals, fish, insects

🐻

Omnivore Eats **both!**
Plants AND meat

💡 **Key vocabulary:** herbivore, carnivore, omnivore, food, survive, sort, plants, meat

📦 Materials

🏠 All from Around the House — No Shopping Needed!

- Toy animals or hand-drawn animal cards (cow, lion, bear, rabbit, eagle, pig, fox, snake, deer, wolf, chicken, raccoon) — or draw animal cards on paper together!
- Three paper plates
- Marker (to label the plates)
- Real food items from the kitchen — lettuce, cheese, nuts, fruit, meat scraps, etc.
- A pet's food bag or label (if you have one!)
- Crayons

🔬 Let's Get Started

Step 1 — The Big Question (3–5 min)
Gather the kids and ask: *"What did YOU eat for breakfast today?"* Let them list things. Then ask: *"Do you think a lion eats the same things as a rabbit? What about a bear?"*
Let them guess — there are no wrong answers yet. Tell them: *"Today we're going to figure out what different animals eat for dinner!"*

Step 2 — Set Up the Plates (3 min)
Label three paper plates with a marker:

- 🌿 **Plate 1:** "Plants Only"
- 🥩 **Plate 2:** "Meat Only"
- 🌿🥩 **Plate 3:** "Both"

Spread them out on the table or floor.

Step 3 — The Sort! (10–15 min)
One at a time, hold up each toy animal or animal card: **cow, lion, bear, rabbit, eagle, pig**.
Ask: *"What does this animal eat? Which plate should it go on?"*
Let kids discuss, debate, and decide together before placing the animal. Celebrate all thinking — even "wrong" guesses lead to great learning.

Step 4 — Kitchen Safari! (5–10 min)
Walk to the kitchen together and look at real food:

- Hold up some lettuce → *"Which animal on our plates would eat this?"*
- Hold up cheese → *"Would a rabbit eat this? Would a lion?"*
- Hold up nuts → *"Could a bear eat this? What about a cow?"*

If you have a pet, look at the food label together — *"What does it eat? Where would your pet go on our plates?"*

Step 5 — Vocabulary Introduction (3–5 min)
Now introduce the science words! Point to each plate as you say:

- 🐰 *"Animals that eat ONLY plants are called **herbivores**!"*
- 🦁 *"Animals that eat ONLY meat are called **carnivores**!"*
- 🐻 *"Animals that eat BOTH plants and meat are called **omnivores**!"*

Ask: *"Which one are YOU? What plate would humans go on?"*

Step 6 — Discussion & Worksheet (5–10 min)
Work through the discussion questions below, then have each child complete their Student Worksheet.

💬 Discussion Questions

🗣️ **Question 1:** "Humans eat both plants and meat. Can you think of another animal that eats both?"

Bears eat both! They catch fish (meat) AND eat berries and honey (plants). Raccoons, pigs, and chickens eat both too. Animals that eat both are called omnivores — just like us!

🗣️ **Question 2:** "If all the rabbits in a forest disappeared, what do you think would happen to the foxes that eat them?"

The foxes would have trouble finding food — they'd need to eat something else or move somewhere new. When one animal disappears, it affects all the animals that depend on it. This is called a food chain!

🚀 Extensions

- 🍽️ **Family Dinner Sort:** Look at your family's dinner tonight and decide — are humans omnivores, herbivores, or carnivores? Which plate would your whole meal go on?
- 🦷 **Teeth Tell the Story:** Look at pictures of animal teeth. Herbivores have flat teeth for grinding plants. Carnivores have sharp teeth for tearing meat. What do humans have?
- 📚 **Read Aloud:** *"What Do Animals Eat?"* or *"Is a Camel a Mammal?"* by Tish Rabe
- 🎨 **Draw a Food Chain:** Draw the sun → grass → rabbit → fox. Each arrow means "gets eaten by." That's a food chain!
- 🌍 **Tricky Cases:** Look up the giant panda — it's technically a carnivore (bear family) but eats mostly bamboo! Nature is full of surprises.

👩‍🏫 Parent/Teacher Notes

- 🥬 **Kitchen Prep Note:** Check your fridge before starting — you'll want at least 2–3 items from the kitchen (e.g., lettuce, apple, cheese). If you don't have these, substitute with any plant food + any animal food. Substitutions are fine — the specific items don't matter, just the category (plant food vs. animal food).
- **All materials from around the house** — no purchases needed. Toy animals work great, but drawing quick animal cards together is also a fun activity!
- **Vocabulary:** Introduce herbivore, carnivore, and omnivore after the sort — let kids discover the categories first, then give them the science words as a "reward."
- **Use real food as props** — lettuce, nuts, a piece of cheese. Making it tangible helps the concept stick at this age.
- **Fun extension:** At dinner, look at your plates and ask "Which category is your meal?" Kids love applying new words to their own lives.
- **Tricky cases to know:** Snakes eat only meat (carnivore). Pandas eat almost only plants (herbivore, even though they're bears). Eagles are carnivores. Bears are omnivores. Pigs and chickens are omnivores.
- **Food chain intro:** The second discussion question gently introduces food chains — a concept they'll explore much more in later lessons. Keep it simple and wonder-filled at this stage.

🍽️ Student Worksheet: What's for Dinner?

Name: _____ Date: _____

Draw each animal in the right column — does it eat plants, meat, or both?

🌿 Eats Plants Herbivore	🥩 Eats Meat Carnivore	🌿🥩 Eats Both Omnivore
Draw animals here	*Draw animals here*	*Draw animals here*

🍽️ Draw Your Favorite Animal's Dinner Plate!

Pick your favorite animal. What would it eat for dinner? Draw the food on its plate!

My favorite animal is: _____ It is a(n): _____ *(herbivore / carnivore / omnivore)*

Draw its dinner here!

✂️ Animal Sort Cards — Cut & Place!

Cut along the dashed lines. Sort each animal onto the correct plate!

🌿 Plants Only | 🥩 Meat Only | 🌿🥩 Both

🐄 **Cow**	🦁 **Lion**	🐻 **Bear**
🐰 **Rabbit**	🦅 **Eagle**	🐷 **Pig**
🦊 **Fox** (mostly carnivore, sometimes omnivore)	🐍 **Snake**	🦌 **Deer**
🐺 **Wolf**	🐓 **Chicken**	🦝 **Raccoon**

🧠 **Bonus challenge!** Can you think of one MORE animal for each plate? Draw it in an empty box above!

🔑 PARENT/TEACHER ANSWER KEY — Detach before giving worksheet to students

🌿 Eats Plants Only *(Herbivores)*	🥩 Eats Meat Only *(Carnivores)*	🌿🥩 Eats Both *(Omnivores)*
🐄 Cow	🦁 Lion	🐻 Bear
🐰 Rabbit	🦅 Eagle	🐷 Pig
🦌 Deer	🐍 Snake	🦊 Fox
	🐺 Wolf	🐓 Chicken
		🦝 Raccoon

💡 Notes on Tricky Animals

- 🐷 **Pig:** Pigs are omnivores! They eat roots, fruit, and small animals. Domestic pigs eat nearly everything — great discussion starter.
- 🦊 **Fox:** Fox classification is genuinely contested — they're primarily carnivores but regularly eat berries, fruit, and insects too. Most biologists classify them as omnivores. *Accept either "carnivore" or "omnivore" for fox, and use it as a teaching moment: "Nature doesn't always fit into neat boxes!"* The sort card is labeled "mostly carnivore, sometimes omnivore" in spirit — both answers are scientifically defensible.
- 🦅 **Eagle:** Eagles eat mostly fish, small mammals, and birds — carnivore. A fun "did you know" for kids.
- 🐓 **Chicken:** Chickens eat seeds AND bugs and worms — they're omnivores! Kids are often surprised by this.
- 🐍 **Snake:** Snakes eat only meat (small animals, eggs). A clear carnivore example.
- 🐻 **Bear:** Bears catch fish AND eat berries, honey, and plants — a classic omnivore and a great reference point for kids.

🗣️ Discussion Question Answers

- **Q1 — Another omnivore?** Bears, raccoons, pigs, and chickens are all omnivores — just like humans! Omnivores can eat a wide variety of foods, which helps them survive in many environments.
- **Q2 — Rabbits disappear?** Foxes (and other rabbit-eaters) would struggle to find food. They'd need to eat something else or move to a new area. This is what scientists call a *food chain* — when one animal disappears, all the animals that depend on it are affected.

✅ What Success Looks Like

- Child can sort at least 8 of the 12 animals correctly.
- Child can explain in their own words what a herbivore, carnivore, and omnivore eat.
- Child can name at least one animal from each category.

- Child can connect their own family's eating to the "omnivore" category.

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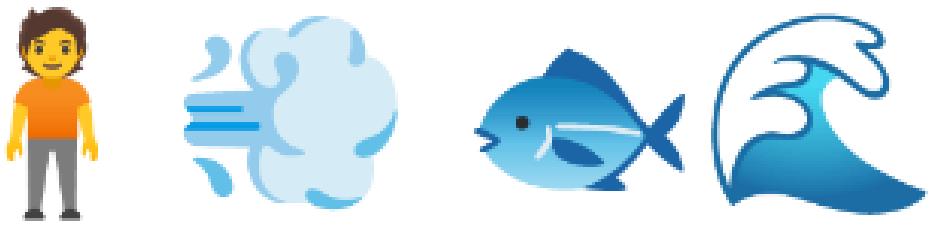
🫁 Breathing: Lungs, Gills, and More 🐟

Life Science — Animal Needs | Unit 3, Lesson 3 of 6

📚 **Grade:** Kindergarten ⏱️ **Duration:** 30–45 minutes 🔬 **Subject:** Life Science — Animal Needs



All animals need oxygen to survive — but not all animals breathe the same way!

📋 **NGSS Standard: K-LS1-1** — Use observations to describe patterns of what plants and animals (including humans) need to survive.

🎯 Learning Objective

Students will **IDENTIFY** that all animals need oxygen and **DESCRIBE** the difference between lungs and gills with at least one example of each.

🔬 The Big Idea

Every animal alive needs **oxygen** — it's one of the most important things on the Animal Survival Checklist! But here's the cool part: animals get oxygen in different ways depending on where they live.

🫁

Lungs

Breathe **air**
Humans, dogs, birds, whales

🐟

Gills

Extract oxygen **from water**
Fish, tadpoles, crabs

🐸

Both!

Gills as babies, lungs as adults
Frogs, salamanders

💡 **Key Vocabulary:** lungs, gills, oxygen, breathe, inhale, exhale

📦 Materials

🏠 All From the Kitchen — No Purchases Needed!

- Clear glass or jar
- Water
- Food coloring (any color)
- Turkey baster *(best tool!)* or a straw for gentle bubbling
- Timer or clock
- Paper and crayons

🔬 Let's Get Started

Step 1 — Feel Your Lungs! (5 min)
Have your child place one hand flat on their chest. Take a big, slow, deep breath in.
"Do you feel that? That's your lungs expanding — filling up with air!"
Now exhale slowly. Feel the chest fall.
Use a timer: **count how many breaths they take in one minute.** Write the number down!
(Typical kindergartner: about 20–30 breaths per minute at rest.)

Step 2 — How Lungs Work (3 min)
Explain: *"When you breathe in, air rushes into your lungs. Your lungs grab the oxygen out of the air and send it into your blood. Then you breathe out the parts your body doesn't need."*
Practice the vocab together: **inhale** (breathe in) and **exhale** (breathe out).

Step 3 — Set Up the Fish Breathing Model (5 min)
Fill the clear glass with water. Add one drop of food coloring and stir gently — the color helps you see the water moving.
"Fish can't breathe air like we do. They breathe water! Their gills pull oxygen right out of the water as it flows through."
Hand your child the turkey baster. Show them how to:

1. Squeeze the bulb and dip it in the water
2. Release slowly — the water is pulled IN (like a fish's mouth opening)
3. Squeeze again — the water is pushed OUT (like water flowing over gills)

Watch the colored water move! *"That flowing water is how a fish gets its oxygen."*

💡 **Straw alternative:** If no turkey baster, use a straw to gently stir and circulate the colored water — this shows how water flows over gills. (Avoid blowing out through the straw, which models exhalation rather than gill function. The goal is to show water *moving* over the gill surface to extract oxygen.)

Step 4 — Compare Side by Side (5 min)
"Let's think about this together."

- Our **lungs** pull oxygen out of **air** → we breathe air in and out
- Fish **gills** pull oxygen out of **water** → water flows in through the mouth and out over the gills

They do the same job — just with different stuff!
Ask: *"Have you ever seen a fish's mouth moving in a fish tank or at the store? That's the gills working!"*

Step 5 — Bonus: Spiracles! (2 min)
*"Here's a wild bonus fact. Insects — like ants, grasshoppers, and beetles — don't have lungs OR gills! They breathe through tiny little holes in the sides of their bodies called **spiracles**. Isn't that amazing?"*
Let them search for a bug outside (or look at a picture) and imagine where those tiny holes might be.

💬 Discussion Questions

🗣️ **Question 1: "Could a fish survive out of water? Why or why not?"**

Give them a moment to think before sharing the hint below.

Suggested Answer: A fish can't survive out of water because its gills only work in water — they need water flowing over them to get oxygen. It's like if we tried to breathe underwater without any equipment — our lungs need air, not water!

🗣️ **Question 2: "Can you think of an animal that can breathe BOTH in water and on land?"**

Hint: Think about an animal that starts life in a pond and grows up to hop on land…

Suggested Answer: Frogs! Tadpoles breathe with gills, but adult frogs breathe with lungs AND can absorb some oxygen right through their damp skin. Turtles and crocodiles also breathe air but spend lots of time in water.

🚀 Extensions

- 🐠 **Fish Tank Watch:** If you have a fish tank at home, observe the fish's mouth and gills moving — that's real breathing!
- 🐸 **Frog Life Cycle:** Draw or look up the frog life cycle. When does it switch from gills to lungs?
- 🔍 **Bug Hunt:** Go outside and find an insect. Explain to it (yes, really!) why its spiracles are cooler than lungs and gills.
- 📚 **Read Aloud:** *"What Do Animals Need?"* or any book about ocean animals and how they breathe.
- 🎨 **Animal Sort:** Draw or find pictures of animals and sort them: lungs, gills, or both!

👩‍🏫 Parent/Teacher Notes

- **All materials from the kitchen** — no purchases needed. The turkey baster is the best tool because pulling and pushing the water is very tactile, but a straw works fine (gently blow into the colored water).
- **Fun connection:** If they've ever seen a fish's mouth moving in a tank or at the store, point out that those are the gills at work — the mouth opens to let water flow in, and it exits over the gills.
- **Don't skip the spiracles!** Kids love the "no lungs, no gills, just holes" fact about insects. It broadens the concept beyond the binary.
- **Frog is the star of the bonus:** If your child is fascinated by frogs, this is a great segue into metamorphosis — coming up later in the unit!
- **Vocab to reinforce:** lungs, gills, oxygen, breathe, inhale, exhale, spiracles
- **If they have a fish tank at home:** this is a perfect time to pause and observe real gills in action before or after the activity.
- **Assessment check:** Can the child explain that all animals need oxygen? Can they name at least two ways animals breathe? That's success!

✏️ Student Worksheet — Breathing

Life Science — Animal Needs | Unit 3, Lesson 3 of 6

Name: _____ Date: _____

🎨 Draw It!

Draw each animal breathing. Use arrows to show the air or water moving!

🧍 A Person Breathing Air

Draw arrows showing air going INTO the lungs

🐟 A Fish Breathing in Water

Draw arrows showing water flowing OVER the gills

📝 Fill It In!

Write or dictate your answers. Use the Word Bank to help!

Lungs breathe _____ .

Gills extract oxygen from _____ .

Word Bank: air | water

🌟 Bonus Question!

Name an animal that can breathe BOTH in water AND on land!

🔢 My Breath Count

How many breaths did you take in one minute?

Write your number in the box!

🔑 PARENT/TEACHER ANSWER KEY — Detach before giving worksheet to student

📝 Fill-In Answers

- **Lungs breathe** → *air*
- **Gills extract oxygen from** → *water*

🌟 Bonus Question

Animals that can breathe in water AND on land:

- 🐸 **Frogs** — tadpoles breathe with gills; adult frogs breathe with lungs and can also absorb some oxygen through their damp skin
- 🐢 **Turtles** — breathe air with lungs but live near/in water
- 🐊 **Crocodiles & Alligators** — breathe air with lungs, swim and hunt in water
- 🦎 **Salamanders** — similar to frogs; some can absorb oxygen through skin

🎨 Drawing Rubric — What to Look For

Drawing	Look For
Person breathing air	Arrows pointing toward nose/mouth or toward chest (lungs); chest area labeled or highlighted
Fish breathing in water	Arrows showing water moving through mouth and/or out the sides (gills); fish shown in water

💬 Discussion Answer Guide

Q1: Could a fish survive out of water?
No — gills only work when water is flowing over them. Out of water, gills can't grab oxygen and the fish can't breathe. Compare: it's like a person trying to breathe underwater — our lungs are built for air, not water.

Q2: An animal that breathes BOTH in water and on land?
Best answer: Frogs. Tadpoles use gills in water → adult frogs use lungs on land and can absorb oxygen through moist skin. Accept any reasonable answer (turtles, crocodiles, salamanders).

✅ What Success Looks Like

- Child can state that all animals need oxygen (air).
- Child can name lungs and gills as two ways animals breathe.
- Child can correctly fill in: lungs breathe air, gills breathe water.
- Child shows arrows on drawings pointing in the correct directions.

- *Bonus:* Child names a frog (or similar) as an animal that can do both.

🔢 Breath Count Reference

Typical resting breath rate for kindergartners: **20–30 breaths per minute**. Any answer in this range is normal and healthy. If they count more after jumping around — great observation to discuss!

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💧 Water, Water Everywhere: How Animals Drink 💧

Life Science — Animal Needs | Unit 3, Lesson 4 of 6

📚 **Grade:** Kindergarten ⏱️ **Duration:** 30–45 minutes 🔬 **Subject:** Life Science — Animal Needs

Every animal needs water — but they don't all get it the same way!

📋 **NGSS Standard:**
K-LS1-1 — Use observations to describe patterns of what plants and animals (including humans) need to survive.

🎯 Learning Objective

Students will **DESCRIBE at least 3 different ways animals get water** and **MATCH** each method to a specific animal example.

💡 The Big Idea

All animals need water to survive — it's one of their four basic needs (remember from Lesson 1!). But not every animal sips from a puddle. Animals have amazing and surprising ways to get the water they need. Dogs lap, frogs soak, and desert tortoises munch on juicy cactus. Today, you'll discover all three methods — and try them yourself!

🔤 **Key Vocabulary:** absorb, moisture, hydration, desert adaptation

absorb **moisture** **hydration** **adaptation** **lap** **permeable**

absorb = to soak up | *moisture* = tiny amounts of water | *hydration* = getting enough water | *adaptation* = a body trick for surviving in tough places | *lap* = to scoop water with the tongue | *permeable* = lets water pass through

📦 Materials

🏠 All Household Items

- Shallow bowls or plates (one per station)
- Water
- Food coloring (for Station 2)
- Paper towels (for Station 2)
- Juicy fruit — orange slices, watermelon chunks, or grapes (for Station 3)
- Towel for cleanup (you'll need it for Station 1! 😄)

🔬 The Science

Water is one of the four basic things every animal needs to survive. But animals have evolved different **adaptations** — clever strategies — to get the water their bodies need:

- **Drinking directly** — like dogs and cats, who lap water from a bowl or stream
- **Absorbing through skin** — like frogs, who can soak up water through their moist skin without even opening their mouths!
- **Eating wet food** — like desert animals that munch on juicy cactus or fruit to get the moisture they need

Each strategy works best in different environments. Desert animals, for example, can't rely on puddles — so they've evolved to get water from their food instead.

> 💡 **Fun facts to share:** Koalas rarely drink water — they get almost all their moisture from eucalyptus leaves! And some desert beetles collect fog on their backs and let it roll down to their mouths. Nature is wild.

🧪 Drinking Stations Activity

Set up three stations and visit each one. Compare which method gets the most water the fastest!

Station 1 — Lap It Up!

Like a dog or cat: Fill a shallow bowl with water. Try lapping water from it using only your tongue — no hands allowed!

Spoiler: It's surprisingly hard for humans. Dogs curl their tongue forward like a ladle scooping forward to scoop water up — try it! (Cats actually curl their tongue backward, which is different. Dogs go forward!)

🧠 **Think about it:** Why would lapping work better than just putting your face in the bowl?

Station 2 — Soak It Up!

Like a frog: Add food coloring to water in a shallow plate so you can see it clearly. Lay a dry paper towel flat on top of the colored water. Watch what happens!

Frogs absorb water directly through their skin — they don't even need to drink! Their skin is specially designed to soak up moisture from the ground, puddles, or damp leaves.

🧠 **Think about it:** Did the paper towel absorb water slowly or quickly? Where do you think frogs like to sit?

Station 3 — Eat It!

Like a desert tortoise: Bite into a juicy piece of fruit — an orange slice, watermelon chunk, or grape. Notice how much liquid is inside!

Many desert animals can't find puddles or streams, so they get their water from the food they eat. Desert tortoises eat juicy cactus pads. Some animals eat insects that are full of moisture.

🧠 **Think about it:** If you lived in a desert with no puddles, which food would give you the most water?

📊 **Compare Results!**
After visiting all three stations, discuss:

- Which method got the most water the fastest?
- Which was the most surprising?
- Which would work best if you lived in a rainy forest? What about a hot, dry desert?

💬 Discussion Questions

🗣️ **Question 1: The Camel Puzzle**

"A camel can go a long time without drinking. Where do you think it stores extra water?"

*Most people think camels store water in their humps — but actually, their humps store fat! Camels can go without drinking for **up to two weeks in cool weather** (less time in heat) because their body is extremely efficient at conserving water — not because they store it. They can drink a huge amount at once (up to 30 gallons!) and then use it very efficiently. Their kidneys and sweat glands are specially adapted to waste as little water as possible.*

🗣️ **Question 2: Desert Life**

"Why do you think animals that live in the desert have to get water from food instead of puddles?"

Deserts get very little rain, so there aren't many puddles or streams. Animals that live there have evolved clever ways to get water from their food — like eating juicy cactus or insects full of moisture. It's called an adaptation — a special trick that helps them survive in a tough environment!

🚀 Extensions

- 🌿 **Koala Research:** Look up photos of koalas eating eucalyptus leaves. They almost never drink water — all their moisture comes from leaves! Can you find another animal like this?
- 🪲 **Fog Beetle:** Draw a picture of a Namib desert beetle collecting fog on its back. Research how its bumpy shell is designed to catch tiny water droplets from morning fog.
- 📚 **Read Aloud:** *"What Do Animals Need?"* or any National Geographic Kids book about desert animals
- 🏜️ **Desert vs. Rainforest:** Draw two habitats side by side. For each one, name 2 animals and show how each gets water.
- 🧪 **Absorption Race:** Test different materials — paper towel vs. cloth vs. plastic wrap. Which absorbs water the fastest? Why do you think frogs have moist, permeable skin?

👩‍🏫 Parent/Teacher Notes

- **All household materials:** Fruit from the kitchen, bowls, paper towels, food coloring — no special supplies needed. **Substitutions are fine** — the specific fruit doesn't matter, just something juicy (wet food category). An orange, grape, watermelon chunk, cucumber, or even a slice of tomato all work.
- **Station 1 warning (in the best way):** The lapping activity is hilarious and kids absolutely love it. Have a towel ready — things will get wet. Let them laugh! That's the fun.
- **Connect to Lesson 1:** Reinforce that water is one of the 4 basic animal needs from earlier in the unit. This lesson shows the *how*, not just the *what*.
- **Fun fact to share:** Koalas get almost all their moisture from eucalyptus leaves and rarely drink water. Some desert beetles collect fog on their backs and let it roll down to their mouths — nature's own water collector!
- **Vocab to reinforce:** *absorb* (to soak up), *moisture* (tiny amounts of water), *hydration* (getting enough water), *desert adaptation* (a special body trick for surviving in dry places)
- **Assessment:** Can the child name at least two different ways animals get water? Can they explain why desert animals eat wet food instead of drinking from puddles? That's success.

💧 Student Worksheet: How Do Animals Get Water? 💧

Name: _____ Date: _____

Draw lines to match each animal with how it gets water!

Draw a line from each number on the left to the matching number on the right.

🐾 Animal	← draw lines here →	💧 How It Gets Water
1. 🐶 Dog		**1.** 🌵 Eats juicy cactus
2. 🐸 Frog		**2.** 💦 Absorbs through skin
3. 🐢 Desert Tortoise		**3.** 🫧 Drinks from puddles

✏️ **Now draw YOUR animal! Pick any animal and show how it gets water.**

My animal: _____ How it gets water: _____

🤔 **Think about it!**

What was your favorite station today? Why?

📋 PARENT/TEACHER ANSWER KEY — Detach before giving worksheet to students

✅ Matching Activity — Correct Answers

Animal	How It Gets Water	Science Note
🐕 **Dog**	🫧 Drinks from puddles	Dogs lap water by curling their tongue **forward** (like a ladle scooping forward) to scoop it up — very efficient! Note: cats curl backward; dogs curl forward.
🐸 **Frog**	💦 Absorbs through skin	Frogs have permeable skin designed to absorb moisture. They often sit in damp soil or shallow water to rehydrate.
🐢 **Desert Tortoise**	🌵 Eats juicy cactus	Desert tortoises eat cactus pads, wildflowers, and other juicy desert plants to get moisture. They can store water in their bladder!

💡 Discussion Answer Notes

Q1 — Camel's hump:

Camels store FAT in their humps, not water! Their amazing bodies are extremely efficient at conserving water. They can drink up to 30 gallons at a time and use it very slowly. Accept any answer that engages with the question — the real learning is in the surprise that it's fat, not water!

Q2 — Desert animals eating wet food:

Deserts get very little rain — sometimes less than 1 inch per year! With no puddles or streams, animals have adapted to get water from juicy plants, insects, and other food sources. Key word to reinforce: adaptation. Look for answers that show understanding of scarcity (not much rain = not much water to drink).

✅ What Success Looks Like

- Child can name at least two different ways animals get water (e.g., drinking and absorbing).
- Child can explain why desert animals eat wet food instead of drinking from puddles (no rain, no puddles).
- Child correctly matches all three animals on the worksheet.
- Bonus: Child uses a vocabulary word like *absorb* or *adaptation* in their explanation.

🐸 Bonus Fun Facts (For Curious Kids!)

- **Koalas** almost never drink water — they get nearly all their moisture from eucalyptus leaves.
- **Namib desert beetles** collect fog on their bumpy backs in the early morning and let tiny droplets roll down to their mouths. A built-in fog collector!

- **Thorny devil lizards** in Australia have tiny channels on their skin that funnel rainwater and morning dew straight to their mouths.
- **Kangaroo rats** in North American deserts almost never drink water — they get all their moisture from dry seeds through a metabolic process.

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🏠 Building a Shelter: Every Animal Needs a Home 🏠

Life Science — Animal Needs | Unit 3, Lesson 5 of 6

📚 **Grade:** Kindergarten ⏱️ **Duration:** 30–45 minutes 🔬 **Subject:** Life Science — Animal Needs
📋 **Unit 3, Lesson 5 of 6**

Birds weave nests, rabbits dig burrows, bears find dens — every animal builds or finds a home!

📋 **NGSS Standard:**
K-LS1-1 — Use observations to describe patterns of what plants and animals (including humans) need to survive.

🎯 Learning Objective

Students will **BUILD a model shelter** for a toy animal using household materials, **IDENTIFY** at least 2 things a shelter must protect an animal from, and **EXPLAIN** why their shelter design meets those needs.

💡 The Big Idea

Every animal needs shelter to survive. A shelter is a safe place that **protects animals from weather** (rain, cold, wind), **hides them from predators** (animals that might eat them), and gives them a cozy place to **rest and raise their babies**. Without shelter, animals would be cold, wet, and in danger.

> 💡 **Key Vocabulary:** shelter, predator, protection, den, burrow, nest, habitat

🔬 The Science

Animals can't just live anywhere — they need a **habitat** that provides food, water, AND shelter. Different animals build or find very different shelters depending on their size, their predators, and the weather in their home environment. A bird's nest is made of twigs and held high in a tree. A rabbit's burrow is dug deep underground. A bear's den might be a cave or a hollow log. A fish might hide inside an underwater cave or rocky crevice. All of these shelters do the same important jobs — they keep the animal safe, dry, and comfortable.

🏠 **Connect to humans:** Our houses are our shelters too! A roof keeps us dry. Walls keep out the wind and cold. Locks keep us safe. We're animals too — we just build fancier homes!

📦 Materials

🧸 You'll Need a Toy Animal

- Any small toy: bird, rabbit, bear, fish, or similar
- A stuffed animal works great too!

🏠 Household Building Supplies

- Cardboard boxes (any size)
- 5–10 small sticks or twigs (pencil-length), collected from outdoors. *No yard? Use toothpicks, wooden coffee stirrers, or pipe cleaners instead!*
- Yarn or string
- Cotton balls
- Towels and blankets
- Pillows
- Tape and scissors
- A bowl or bin (for fish option)
- A spray bottle with water (for shelter testing!)

🔬 Let's Get Started

Step 1 — What Is a Shelter? (5 min)
Start with a question: *"What does a bird need to survive? What about a bear? What about a rabbit?"*

After students name food and water, ask: *"What about a safe place to sleep? A place to have babies? A place to hide from danger? That's called a SHELTER."*

Show pictures or draw simple sketches of: a bird nest, a rabbit burrow, a bear den, a fish cave. Let kids identify them.

Step 2 — Choose Your Animal (3 min)

Each student picks a toy animal and decides what kind of shelter it needs:

Bird → Nest

Use sticks/twigs from the yard, yarn, and cotton balls in a bowl

Rabbit → Burrow

Drape towels over a cardboard box with a "tunnel" entrance

Bear → Den

Stack blankets and pillows into a cozy cave shape

Fish → Underwater Hiding Spot

Fish find shelter in underwater plants and rocks! Create a safe hiding spot for a toy fish using a cup and crumpled paper (rocks) with a few sprigs of fake or real plant tucked around it. ***Note: this is a dry, tabletop model — no water needed! We're representing what an underwater shelter would look like.***

Step 3 — Build the Shelter! (10–15 min)

Using ONLY household materials, build the shelter for the chosen animal. Encourage kids to:

- Think about the SIZE — can the animal fit inside?
- Think about COVERING — is there a roof or walls?
- Think about HIDING — can the animal be hidden from outside?
- Add a space for "babies" — make it big enough for a family!

"Real animals are incredible engineers. You are too!"

Step 4 — Test the Shelter! (5–10 min)
Now it's time to see if the shelter really works. Run each test:

🧪 Shelter Test Checklist

- [ ] 🌧️ **"Rain" test:** Lightly sprinkle water near the shelter with a spray bottle. Does the animal stay dry?
- [ ] 🐣 **"Baby" test:** Could another small toy animal fit inside? Is there room for a family?
- [ ] 👁️ **"Predator" test:** Step back and look — can you see the animal from outside? Is it hidden?

If a test fails, IMPROVE the shelter! Real animals do this too — they adjust and rebuild.

Step 5 — Share & Discuss (5 min)
Each child presents their shelter. Ask: *"What did you build? What does it protect your animal from? What would you change to make it even better?"*

💬 Discussion Questions

🗣️ **Question 1:** "What would happen if a bird didn't have a nest?"

Without a nest, a bird's eggs would fall to the ground and break, and baby birds would be exposed to rain, cold, and predators. The nest keeps the family safe, warm, and together — it's like a nursery and a fortress!

🗣️ **Question 2:** "Humans build houses for shelter. What's one thing a human house does that an animal's shelter also needs to do?"

Both need to keep out rain, cold, and wind — provide safety from danger — and have space for the family! Animals are clever builders: beavers build dams, birds weave nests, some fish dig holes in sand. We're not so different from animals after all!

🚀 Extensions

- 🦫 **Beaver Dam Challenge:** Can you build a dam with sticks across a shallow tray of water? Will it hold? Beavers do this to make a safe pond for their lodge!
- 🕸️ **Spider Web Hunt:** Go outside and find a spider web. What materials did the spider use? How is it like a shelter?
- 📚 **Read Aloud:** *"A House Is a House for Me"* by Mary Ann Hoberman — a classic exploration of all kinds of animal homes!
- 🗺️ **Animal Home Map:** Draw a neighborhood — but for animals! Where does the bird live? The rabbit? The bear? Label each home.
- 🏘️ **Real vs. Model:** Look up pictures of real bird nests, beaver dams, and termite mounds. How close did your model get?

👩‍🏫 Parent/Teacher Notes

- **All household materials — nothing to buy.** Sticks and twigs from the yard are perfect for bird nests. Raid the blanket closet for bear dens!
- **This is an engineering activity as well as science.** When students test their shelters and find they fail, encourage them to iterate and improve — that's real science and engineering thinking. Failure is part of the process!
- **Connect back to humans:** Our houses are our "shelters" too — what does our house protect us from? Rain, wind, cold, danger? Help kids see that humans and animals have the same survival needs.
- **Discuss different real animal shelters:** birds' nests, beaver dams, rabbit burrows, spider webs, termite mounds, prairie dog towns. Each one is perfectly designed for that animal's needs.
- **Vocab to reinforce:** shelter, predator, protection, den, burrow, nest, habitat
- **Assessment tip:** Can the child name at least two things a shelter protects an animal from? Can they explain why their shelter works (or what it needs to work better)? That's the learning goal.

🏠 My Animal Shelter — Student Worksheet

Name: _____ Date: _____

🐾 My animal is: _____

🏠 The type of shelter I built: _____

> 📝 **Parent/Scribe Note:** For Task 2, help your child label their drawing by writing the material names for them as they point and tell you what each part is. Kindergartners can dictate — you can be their pencil!

✏️ Task 1 — Draw your shelter!

Draw your shelter with your animal inside. Show what the shelter looks like from the outside.

🏷️ Task 2 — Label 3 materials you used:

Add arrows to your drawing above and write (or have a grown-up write) the name of each material.

_____ _____ _____

💬 Task 3 — Fill in the sentence:

My shelter protects my animal from _____.

✂️ PARENT/TEACHER ANSWER KEY — Detach before giving worksheet to students

💬 Discussion Question 1

Q: "What would happen if a bird didn't have a nest?"

A: Without a nest, a bird's eggs would fall to the ground and break, and baby birds would be exposed to rain, cold, and predators. The nest keeps the family safe, warm, and together — it's like a nursery and a fortress!

💬 Discussion Question 2

Q: "Humans build houses for shelter. What's one thing a human house does that an animal's shelter also needs to do?"

A: Both need to keep out rain/cold/wind, provide safety from danger, and have space for the family. Animals are incredibly clever builders — beavers build dams, birds weave nests, some fish dig holes in sand. Humans and animals share the same core survival needs!

✏️ Worksheet Fill-In

"My shelter protects my animal from ___."
Accept any of the following (great answers include multiple): rain, cold, wind, predators, danger, getting wet, the cold ground, being found by enemies, bad weather. A child who names 2 or more is demonstrating excellent understanding.

🧪 Shelter Test Expectations

- 🌧️ **Rain test:** The shelter should provide cover. If it doesn't, guide the student to add a "roof" layer (a box lid, extra towel, or leaves).
- 🐣 **Baby test:** If the shelter is too small, encourage expanding it. Real animals often enlarge their nests/burrows as their family grows!
- 👁️ **Predator test:** If the animal is visible, encourage adding walls or a "door." Rabbits' burrow entrances are often disguised with leaves and sticks.

✅ What Success Looks Like

- Student built a shelter that covers/hides the animal in some way
- Student can name at least 2 things a shelter protects an animal from
- Student can explain the function of their shelter (not just describe what it looks like)
- Bonus: Student made at least one improvement after a failed test

🐾 Real-World Animal Shelter Reference

- 🐦 **Bird nest:** Woven from twigs, grass, mud — keeps eggs warm and off the ground
- 🐇 **Rabbit burrow:** Tunnels underground — warm, hidden from foxes and hawks

- 🐻 **Bear den:** Cave, hollow log, or dug-out space — for hibernating through winter
- 🐟 **Fish hiding spot:** Underwater plants and rocks — hides from larger fish predators. Model: cup + crumpled paper (rocks) + plant sprigs
- 🦫 **Beaver lodge:** Built from sticks and mud in the middle of a pond — entrance is underwater for protection!
- 🕷️ **Spider web:** A trap AND a home — some spiders hide in curled leaves attached to their web
- 🐜 **Termite mound:** Can be taller than a human — temperature-regulated inside!

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🌿 When Needs Aren't Met: What Happens? 🌿

Life Science — Animal Needs | Unit 3, Lesson 6 of 6

📚 **Grade:** Kindergarten ⏱️ **Duration:** 30–45 minutes 🔬 **Subject:** Life Science — Animal Needs
📋 **Unit:** Unit 3, Lesson 6 of 6

Birds migrate, bears hibernate, squirrels adapt — every animal has a plan for survival!

📋 **NGSS Standard:**
K-LS1-1 — Use observations to describe patterns of what plants and animals (including humans) need to survive.

🎯 Learning Objective

Students will **PREDICT and EXPLAIN** what happens when an animal can't get one of its basic needs, and **NAME** at least 2 of the 3 survival strategies (migrate, hibernate, adapt) with a real animal example for each.

💡 **Key Vocabulary:**

migrate **hibernate** **adapt** **survive** **predator** **prey** **food chain**

🔬 The Big Idea

We've learned that every animal needs **food, water, air, and shelter** to survive. But what happens when one of those needs suddenly disappears? Animals can't just wait and hope — they have to *do something*. In nature, animals have three main survival strategies:

🐦

Migrate

Move to a new place where needs CAN be met. Birds fly south in winter to find

🐻

Hibernate (or Torpor)

Go into a deep sleep and use stored energy to survive until conditions

🐿️

Adapt

Change behavior or grow special features — like squirrels storing nuts, or

food and warmth.

improve. Bears actually enter *torpor* — a lighter sleep. Ground squirrels and bats do true hibernation!

animals growing thicker fur.

📦 Materials

🏠 Gather These Household Items

- Toy animals (any mix — farm animals, wildlife, stuffed animals all work!)
- Blue fabric or blue paper (to represent a pond or water source)
- Green fabric or green paper (to represent plants, meadow, or forest)
- One large sheet of paper (butcher paper, poster board, or taped-together sheets)
- Crayons or markers to draw and label the landscape

✅ *No special supplies needed — this is a storytelling + dramatic play lesson!*

🔬 Let's Get Started

Step 1 — Build the Landscape (5 min)
Spread a large sheet of paper on the floor or table. Together, draw and set up three habitats:

- 🔵 **A pond** — lay down the blue fabric/paper. Place some water animals here (duck, frog, fish).
- 🟢 **A meadow/forest** — lay down the green fabric/paper. Add plant-eating animals (rabbit, deer, caterpillar).
- 🌄 **Open land** — the rest of the paper. Place meat-eaters here (fox, owl, hawk).

Say: *"This is our animal world. Every animal here has everything it needs right now — food, water, air, and shelter. Let's see what happens if things change…"*

Step 2 — Play "What If?" Scenarios (15 min)
Act out each scenario below. Let your child move the toy animals around as you discuss. Encourage them to narrate what the animals are thinking and doing!

🌊 Scenario 1: "What if the pond dries up?"

→ Animals that need water must leave! Move the duck and frog away from the dry pond. Where will they go? They'll need to find a new water source. Some might not make it if there's no water nearby.

🌿 Scenario 2: "What if all the plants disappear?"

→ Plant-eaters have no food — they must move to find more plants! Move the rabbits and deer away. But wait… the foxes who eat rabbits now have NO food either. Remove the food → the whole chain is affected! (Simple food chain intro!)

🦁 Scenario 3: "What if a dangerous new predator moves in?"

→ Prey animals need to hide or move away to stay safe! Their shelter need is threatened. Some might find new hiding spots; others might need to leave the area entirely.

Step 3 — Connect to Real Life (5–8 min)
Ask these one at a time and let your child think and answer:

- 🐦 *"What do birds do in winter when food is scarce?"* → **Migrate!** They fly south where it's warmer and food is easier to find.
- 🐻 *"What do bears do when winter comes?"* → **Hibernate!** They eat a LOT of food beforehand, then sleep all winter using that stored energy.
- 🐿️ *"What do squirrels do to prepare for winter?"* → **Adapt their behavior!** They collect and hide nuts all fall so they have food stored when winter arrives.

Step 4 — Discussion & Worksheet (10 min)
Discuss the questions below, then complete the Student Worksheet together (page 3). Students draw each scenario and show how the animal solves its problem.

💬 Discussion Questions

🗣️ **Discussion Question 1:**
"In winter, it gets very cold and food is hard to find. What are three things animals might do?"

Suggested answer: Animals have three main strategies: (1) Migrate — move to a warmer place with more food, like birds flying south. (2) Hibernate — go into a deep sleep and use stored energy, like bears. (3) Adapt — grow thicker fur, store food ahead of time, like squirrels hiding nuts.

🗣️ **Discussion Question 2:**
"If you were an animal, which survival strategy would you choose — migrate, hibernate, or adapt? Why?"

No wrong answer here! Look for creative reasoning: migrate (want to travel, go somewhere warm and exciting), hibernate (like to sleep, want to save energy), adapt (want to stay home, be clever about it). All are valid real strategies animals actually use!

🚀 Extensions

- 🦅 **Migration Map:** Look up where monarch butterflies migrate. Trace the path on a U.S. map together — it goes all the way to Mexico!
- 🐻 **Hibernation Science:** Look up what animals actually do during hibernation — their heart rate slows way down! Compare it to your resting heart rate.
- 📚 **Read Aloud:** *"A Warmer World"* by Caroline Arnold, or *"Animals in Winter"* by Henrietta Bancroft
- 🌍 **Food Chain Challenge:** Draw a simple food chain together: grass → rabbit → fox. What happens if the rabbit disappears? What if the grass disappears?
- 🎭 **Animal Theater:** Act it out! One person is the narrator, others are animals. Play out a drought scenario — who survives, who has to move?

🏆 Unit 3 Wrap-Up — The Survival Checklist

Congratulations — you've completed **all 6 lessons** in Unit 3: Animal Needs! Here's everything you've learned:

🐾

Lesson 1 — What Animals Need
All animals share four basic needs: food, water, air, and shelter.

🍽️

Lesson 2 — What Animals Eat
Animals eat plants (herbivores), meat (carnivores), or both (omnivores).

🫁

Lesson 3 — How Animals Breathe
Animals get oxygen through lungs, gills, or even skin — all need oxygen!

💧

Lesson 4 — How Animals Drink
Animals drink, absorb through skin, or eat wet food to get water.

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Lesson 5 — Building Shelter
Animals build nests, burrows, and dens to stay safe, warm, and dry.

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Lesson 6 — When Needs Aren't Met
When needs aren't met, animals migrate, hibernate (or enter torpor), or adapt.

🐾 Culminating Activity: The Survival Checklist Poster

⏱️ **Time:** 30–45 minutes 🖍️ **Materials:** Large paper (poster board or taped sheets), crayons or markers

Pick a favorite animal and create a poster showing how it meets ALL 4 of its basic needs!

1. Draw your animal in the center of the poster.
2. Add 4 sections around it — one for each need: 🍎 Food, 💧 Water, 💨 Air, 🏠 Shelter.
3. In each section, draw and write how YOUR animal gets that need met. Be specific!
4. Add a bonus section: *"What does this animal do when things go wrong?"* (migrate, hibernate, or adapt?)
5. **Present your poster to a family member!** Explain each need and how your animal survives.

Any animal works — a hummingbird, a polar bear, an octopus, a horse! The more interesting, the better.

👩‍🏫 Parent/Teacher Notes

- 🐻 **Hibernate vs. Torpor:** True hibernation means body temperature drops dramatically — ground squirrels and bats do this. Bears actually enter *torpor* — a lighter sleep where they can wake up more easily. At Kindergarten level, "hibernate" is perfectly acceptable! But if your child asks why bears wake up mid-winter, this is the accurate answer: bears are in torpor, not true hibernation.
- **All household materials:** Toy animals + paper + crayons is everything you need. No prep required.
- **This is a storytelling lesson:** Let the child narrate what the animals are doing and thinking. Dramatic play is genuine science learning at this age — don't rush it.
- **Tie back to every previous lesson:** This is the culminating lesson! Reference the four needs from Lesson 1 (food, water, air, shelter), how animals eat from Lesson 2, breathing/oxygen from Lesson 3, getting water from Lesson 4, and shelter-building from Lesson 5 as you play through scenarios. "Remember when we learned about water needs in Lesson 4? What happens to animals that need the pond?"
- **Simple food chain intro:** Don't over-complicate it. Just: "Rabbits eat grass. Foxes eat rabbits. If the rabbits are gone, what happens to the foxes?" That's enough for Kindergarten.
- **Vocabulary tip:** Write migrate, hibernate, and adapt on sticky notes and act them out physically — "migrate" means walk across the room; "hibernate" means pretend to sleep. Body memory sticks!
- **Culminating poster:** This can be done over several days. Let the child research their animal if they want — a great chance to use books or kid-safe websites together.
- **Presenting to a family member:** This is powerful. The act of teaching someone else cements learning. Even "presenting" to a stuffed animal counts for shy kids!

🌿 Student Worksheet — When Needs Aren't Met

Name: _____ Date: _____

For each scenario, draw the **problem** AND the animal's **solution**. What does the animal do to survive?

(Parent note: You can be your child's scribe — let them tell you what to write!)

💧 Scenario 1: The water source dries up!

An animal that lives near a pond wakes up one morning and the pond is completely gone. What happens?

🚨 Draw the PROBLEM:

Draw here!

✅ Draw the SOLUTION:

Draw here!

What does the animal do? _____

🍎 Scenario 2: All the food disappears!

A rabbit wakes up and all the plants and grass are gone. There is nothing left to eat.

🚨 Draw the PROBLEM:

Draw here!

✅ Draw the SOLUTION:

Draw here!

What does the animal do? _____

🌟 Bonus — Circle your survival strategy!

If YOU were an animal facing a tough winter, which strategy would you pick?

🐦	🐻	🐿️
Migrate	**Hibernate**	**Adapt**
Move somewhere new	Sleep through it	Get clever & prepare

Why? _____

🔑 PARENT/TEACHER ANSWER KEY — Detach before giving worksheet to students

💧 Scenario 1: Water source dries up

Problem: The pond is gone — animal has no water nearby.
Expected responses (any of these are correct):

- The animal **migrates** — moves to find a new pond, river, or water source.
- Some animals (like camels) can **adapt** by storing water in their bodies, or finding water in food.
- If the animal can't find water, it won't survive — this is a valid and honest scientific answer too.

Look for: the animal moving toward a new water source, OR drawing the animal in distress without water. Both show understanding.

🍎 Scenario 2: Food source disappears

Problem: Rabbit has no plants to eat.
Expected responses:

- The rabbit **migrates** to a new area where plants are growing.
- The rabbit might **adapt** by eating a different type of food if available.
- Bonus understanding: *If the rabbit leaves, the fox that eats rabbits ALSO has no food* — this is the food chain concept. Celebrate this if the child mentions it!

Look for: the rabbit moving away, OR showing the ripple effect on predators. Either demonstrates strong understanding.

🏠 Scenario 3: Shelter is destroyed ⭐ Extension Only (no corresponding worksheet question)

> ***Note for educators:*** *The Student Worksheet only includes Scenario 1 (water) and Scenario 2 (food). Use this scenario as an optional extension discussion — it does not have a corresponding worksheet drawing prompt.*

Problem: Bird's nest is gone — no safe place to rest or reproduce.
Expected responses:

- The bird **adapts** — builds a new nest! This is very common and shows resourcefulness.
- The bird **migrates** — finds a safer tree or area with better shelter.
- The bird finds an existing hollow in a tree, a cliff face, or another natural shelter structure.

Look for: the bird building a new nest OR flying to find a new home. Either is correct.

🌟 Bonus — Strategy Choice

This is open-ended — **there is no wrong answer.** Evaluate based on reasoning quality:

- **Migrate:** Valid if child says they want to go somewhere warmer, find more food, explore a new place.

- **Hibernate:** Valid if child says they like sleeping, want to save energy, or wait for things to get better.
- **Adapt:** Valid if child says they want to stay home, be clever, prepare ahead, or change what they eat/wear.

All three are real strategies real animals use. Celebrate creative reasoning!

✅ What Success Looks Like — End of Unit 3

- Child can name all 4 basic animal needs: food, water, air, shelter.
- Child can describe what happens when one need isn't met (animal must move or adapt).
- Child can name and explain at least 2 of the 3 survival strategies: migrate, hibernate, adapt.
- Child can give a real animal example for each strategy (bird → migrate, bear → hibernate, squirrel → adapt).
- *Bonus:* Child understands that food chains connect animals — losing one affects others.

🌟 Culminating Project Rubric — Survival Checklist Poster

Element	Developing ⭐	Proficient ⭐⭐	Excellent ⭐⭐⭐
Animal chosen & drawn	Animal is named	Animal is drawn with details	Animal drawn with habitat clues
All 4 needs addressed	2 needs shown	All 4 needs shown	All 4 needs with specific details
Survival strategy identified	Strategy named	Strategy named + explained	Strategy with real-life example
Presentation to family	Describes poster briefly	Explains each need	Explains + answers questions

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